

# Jack characters as generating series of bipartite maps and proof of Lassalle's conjecture

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# Symmetric functions

$\lambda = [\lambda_1, \lambda_2, \dots, \lambda_k]$  is a partition of  $n$  if  $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k$  and  $\lambda_1 + \lambda_2 + \dots + \lambda_k = n$ . The integer  $n = |\lambda|$  is the size of  $\lambda$  and  $k = \ell(\lambda)$  is its length.

We consider the space of symmetric functions on an alphabet  $\mathbf{x} := x_1, x_2, \dots$

For  $m \geq 0$ , the power-sum function  $p_m$  is defined by

$$p_m(\mathbf{x}) := \sum_{i \geq 1} x_i^m,$$

and if  $\lambda = [\lambda_1, \lambda_2, \dots, \lambda_k]$  then

$$p_\lambda(\mathbf{x}) = p_{\lambda_1}(\mathbf{x}) p_{\lambda_2}(\mathbf{x}) \cdots p_{\lambda_k}(\mathbf{x}).$$

# Symmetric functions

Jack polynomials  $J_\lambda^{(\alpha)}$ ; are symmetric functions which depend on a parameter  $\alpha$ . They can be characterized as the unique family of symmetric functions which satisfies;

- *Triangularity:*

$$J_\lambda^{(\alpha)} = \sum_{\mu \leq \lambda} a_{\lambda, \mu}(\alpha) m_\mu$$

- *Normalization:*

$$[p_{1^{|\lambda|}}] J_\lambda^{(\alpha)} = 1$$

- *Orthogonality:*

$$\langle J_\lambda^{(\alpha)}, J_\mu^{(\alpha)} \rangle_\alpha = 0 \text{ if } \lambda \neq \mu,$$

where

$$\langle p_\lambda, p_\mu \rangle_\alpha = \delta_{\lambda, \mu} \alpha^{\ell(\lambda)} z_\lambda.$$

- When we take  $\alpha = 1$ , we obtain Schur functions.
- They can be obtained from Macdonald polynomials  $J_\lambda^{(q, t)}$  by taking  $q = t^\alpha$  and the limit  $t \rightarrow 1$ .

# Main result

Several combinatorial formulas for Schur functions

- **Tableaux**, generalized to Jack and Macdonald polynomials.
- **Maps**, conjectures for Jack polynomials (Hanlon '88, Goulden–Jackson '96).

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## Main result

*A combinatorial interpretation in terms **maps** of the power-sum expansion of Jack polynomials, which we deduce from a more general formula for **Jack characters**.*

- gives a topological expansion for Jack characters, generalizing a known formula for the characters of the symmetric group in terms of pairs of permutations/orientable maps.
- gives an answer in some sense to a conjecture of Hanlon 1988.
- answers a positivity conjecture of Lassalle 2008 about Jack characters.

# Jack characters

## Definition (Kerov–Olshanski, Lassalle)

Fix a partition  $\mu$ .

$$\theta_{\mu}^{(\alpha)}(\lambda) := \begin{cases} 0, & \text{if } |\lambda| < |\mu|. \\ \binom{|\lambda| - |\mu| + m_1(\mu)}{m_1(\mu)} [p_{\mu, 1^{|\lambda| - |\mu|}}] J_{\lambda}^{(\alpha)}, & \text{if } |\lambda| \geq |\mu|. \end{cases}$$

where  $m_1(\mu)$  is the number of parts of size 1 in  $\mu$ . In particular,

$$J_{\lambda}^{(\alpha)} = \sum_{|\mu|=|\lambda|} \theta_{\mu}^{(\alpha)}(\lambda) p_{\mu}.$$

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**Recall:**

$$s_{\lambda} = \sum_{|\mu|=|\lambda|} \frac{\chi^{\lambda}(\mu)}{z_{\mu}} p_{\mu} \quad \text{where } z_{\lambda} := \frac{|\lambda|!}{|\mathcal{C}_{\lambda}|}.$$

For  $\alpha = 1$ ,  $\theta_{\mu}^{(1)}(\lambda)$  correspond to the characters of the symmetric group.



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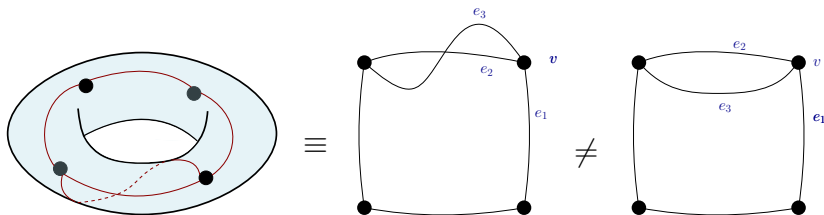
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For  $\alpha = 1$ ,  $\theta_{\mu}^{(1)}(\lambda)$  correspond to the characters of the symmetric group.

For  $\alpha = 2$ ,  $\theta_{\mu}^{(2)}(\lambda)$  correspond to zonal characters.

# Maps

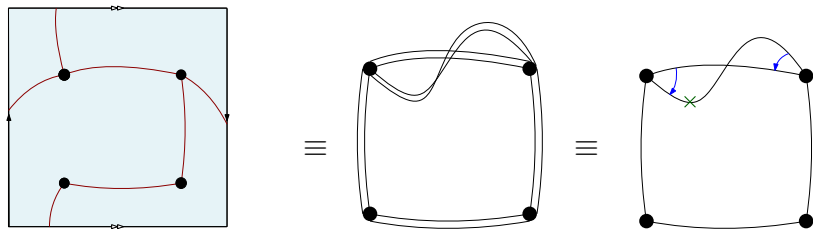
- A *connected map* is a cellular embedding of a connected graph in a surface, **orientable or not**.



Orientable connected maps

# Maps

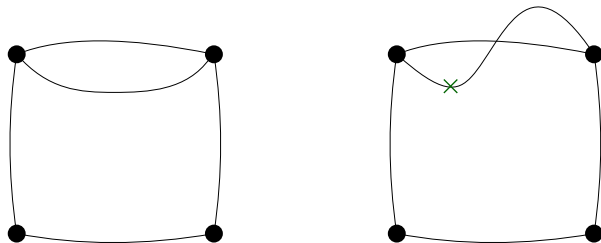
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A non orientable connected map

# Maps

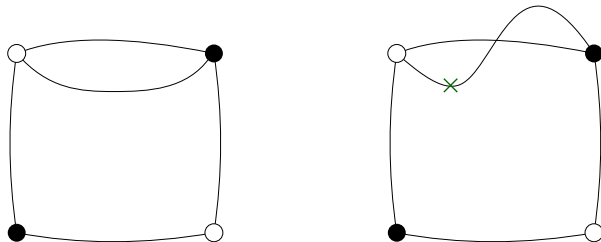
- A *connected map* is a cellular embedding of a connected graph in a surface, **orientable or not**.
- A map is a collection of connected maps. A map is *orientable* if its all connected components are embedded into orientable surfaces.



A non orientable map

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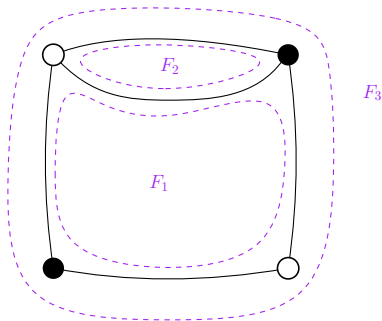
- A *connected map* is a cellular embedding of a connected graph in a surface, **orientable or not**.
- A map is a collection of connected maps. A map is *orientable* if its all connected components are embedded into orientable surfaces.
- A map is bipartite if its vertices are colored in white and black, and each white vertex has only black neighbors.



A non orientable bipartite map

# Maps

- The size of a map is its number of edges.
- The **face partition** of a bipartite map  $M$ , denoted by  $\diamond(M)$ , is the partition given by the face degrees, divided by 2.



A map of size 5 and face partition  $[2, 2, 1]$ .

# Maps

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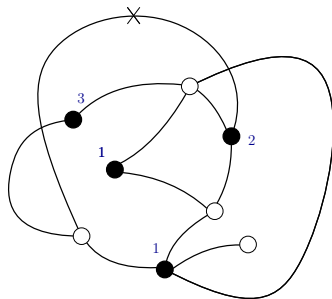
We associate to a face of degree  $2r$ , a weight  $p_r$ .

→ To a map  $M$ , we associate the face-weight  $p_{\diamond(M)}$ .

# Layered maps

Let  $k$  be a positive integer. A map  $M$  is  $k$ -layered if

- each black vertex has a label in  $1, 2, \dots, k$ .



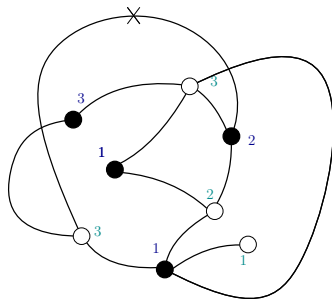
A 3-layered map



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- each white vertex is labelled by the maximal label among the labels of its black neighbors.



A 3-layered map

## Case $\alpha = 1$ : Irreducible characters of the symmetric group

### Theorem (Stanley–Féray formula, Féray '10)

$$\theta_{\mu}^{(1)}(\lambda) = \sum_{\substack{\text{orientable maps } M \\ \ell(\lambda)\text{-layered of face partition } \mu}} \frac{(-1)^{|\mu|}}{z_{\diamond(M)}} \prod_{1 \leq i \leq \ell(\lambda)} (-\lambda_i)^{|\mathcal{V}_{\circ}^{(i)}(M)|},$$

where  $|\mathcal{V}_{\circ}^{(i)}(M)|$  is the number of white vertices labelled by  $i$ .

→ to a white vertex in layer  $i$  we associate the weight  $(-\lambda_i)$ .

#### Proof :

- Tools from representation theory of  $\mathfrak{S}_n$  (Young symmetrizers).
- Encoding of orientable maps with  $n$  edges with pairs of permutations  $\mathfrak{S}_n$ .

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where  $|\mathcal{V}_{\circ}^{(i)}(M)|$  is the number of white vertices labelled by  $i$ .

## Case $\alpha = 2$ : Zonal characters

### Theorem (Féray–Śniady '11)

$$\theta_{\mu}^{(2)}(\lambda) = \sum_{\substack{\text{maps } M \text{ orientable or not} \\ \ell(\lambda)\text{-layered of face partition } \mu}} \frac{(-1)^{|\mu|}}{2^{\ell(\diamond(M))} z_{\diamond(M)}} \prod_{1 \leq i \leq \ell(\lambda)} (-2\lambda_i)^{|\mathcal{V}_{\circ}^{(i)}(M)|}.$$

# Statistics of non-orientability

## Definition (Goulden–Jackson '96)

A **statistic of non-orientability** (on  $k$ -layered maps) is a statistic which associates to each  $k$ -layered map  $M$  a non-negative integer such that  $\vartheta(M) = 0$  if and only if  $M$  is orientable.

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When  $\alpha = 1$ :

$$b^{\vartheta(M)} = \begin{cases} 1 & \text{if } M \text{ is orientable} \\ 0 & \text{otherwise.} \end{cases}$$

When  $\alpha = 2$ :

$$b^{\vartheta(M)} = 1 \quad \text{for any } M.$$

# Main result

## Theorem (BD–Dołęga '23+)

*There exists a statistic of non-orientability  $\vartheta$  on layered maps, such that*

$$\theta_{\mu}^{(\alpha)}(\lambda) = (-1)^{|\mu|} \sum_{\substack{\text{layered maps } M \\ \text{of face partition } \mu}} \frac{b^{\vartheta(M)}}{2^{|\mathcal{V}_{\bullet}(M)| - cc(M)} \alpha^{cc(M)}} \prod_{i \geq 1} \frac{(-\alpha \lambda_i)^{|\mathcal{V}_{\circ}^{(i)}(M)|}}{z_{\nu_{\bullet}^{(i)}(M)}}, \quad (1)$$

- $|\mathcal{V}_{\bullet}(M)|$  is the number of black vertices of  $M$ ,
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- For  $\alpha = 1$ : Stanley–Féray formula for the characters of the symmetric group.
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# Jack polynomials in the power-sum basis

## Theorem (BD–Dołęga '23+)

*There exists an explicit statistic of non-orientability  $\vartheta$ , such that*

$$J_{\lambda}^{(\alpha)} = (-1)^{|\lambda|} \sum_{\substack{\ell(\lambda) \text{--} \text{layered} \\ \text{maps } M}} \frac{p_{\diamond(M)} b^{\vartheta(M)}}{2^{|\mathcal{V}_{\bullet}(M)| - cc(M)} \alpha^{cc(M)}} \prod_{1 \leq i \leq \ell(\lambda)} \frac{(-\alpha \lambda_i)^{|\mathcal{V}_{\circ}^{(i)}(M)|}}{z_{\nu_{\bullet}^{(i)}(M)}},$$

- $p_{\diamond(M)}$  is the power-sum function associated to the partition  $\diamond(M)$ ,
- $|\mathcal{V}_{\bullet}(M)|$  is the number of black vertices of  $M$ ,
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- $cc(M)$  is the number of connected components of  $M$ .

- a face-weight  $p_{\diamond(M)}$ ,
- a non-orientability weight  $b^{\vartheta(M)}$ ,
- a weight related to layer structure  $\prod_{1 \leq i \leq \ell(\lambda)} (-\alpha \lambda_i)^{|\mathcal{V}_{\circ}^{(i)}(M)|}$

# How to obtain statistics of non-orientability?

**General method** (La Croix '09, Dołęga–Féray–Śniady '14, Chapuy–Dołęga'22).

- 1 Fix a map  $M$  of size  $n$ . We choose a **a decomposition algorithm**, by fixing a total order on the edges of  $M$ :  $e_1, e_2, \dots, e_n$ . We denote  $M_i := M \setminus \{e_1, e_2, \dots, e_{i-1}\}$ .

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- 2 For any map  $M$  and edge  $e$ , we choose

$$\vartheta(M, e) \in \{0, 1\}.$$

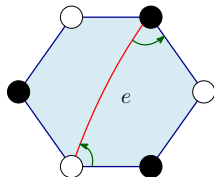
We then set

$$\vartheta(M) = \sum_{1 \leq i \leq n} \vartheta(M_i, e_i).$$

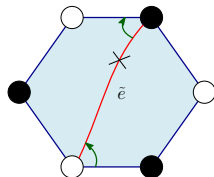
# Edge weights

Let  $M$  be a map. We fix two corners of  $M$  of different colors and we connect them by an edge. We distinguish many cases.

- 1 The two corners are in the same face of  $M$



a border  
 $\vartheta(M \cup \{e\}, e) = 0$

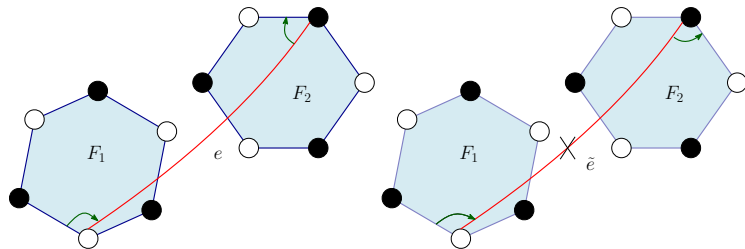


a twist  
 $\vartheta(M \cup \{e\}, e) = 1$

# Edge weights

Let  $M$  be a map. We fix two corners of  $M$  of different colors and we connect them by an edge. We distinguish many cases.

- 2 The two corners are in two different faces of the same connected component.



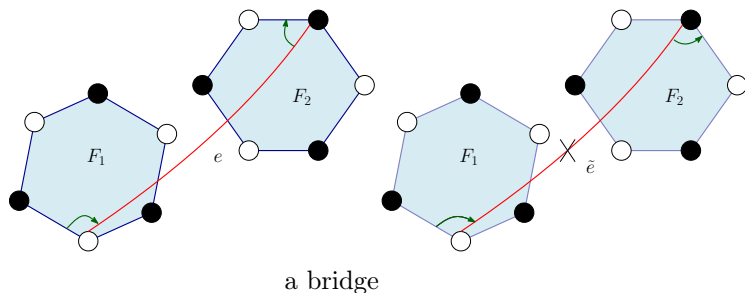
a handle

- ▶  $\vartheta(M \cup \{e\}, e) + \vartheta(M \cup \{\tilde{e}\}, \tilde{e}) = 1$ .
- ▶ if  $M$  is orientable then exactly one of the maps  $(M \cup \{e\}, e)$  and  $(M \cup \{\tilde{e}\}, \tilde{e})$  is orientable, we associate to it the weight 0, and to the other map the weight 1.

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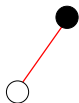
- 3 The two corners are in two different connected components.



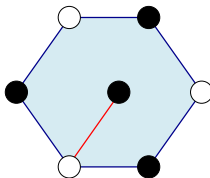
►  $\vartheta(M \cup \{e\}, e) = \vartheta(M \cup \{\tilde{e}\}, \tilde{e}) = 0.$

# Edge weights

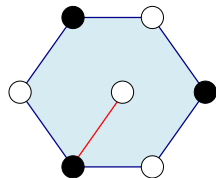
- 1 the added edge is connected to a new vertex  
 $\longrightarrow \vartheta(M \cup \{e\}, e) = 0.$



an isolated edge

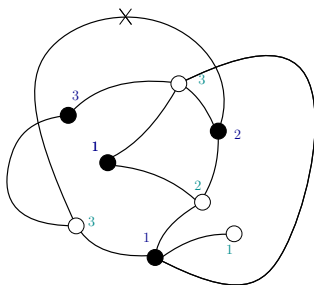


a black leaf



a white leaf

# How to decompose a layered map?



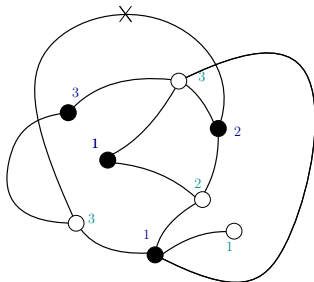
A non-orientable 3-layered map

Recall:

- each black vertex has a label in  $1, 2, \dots, k$ .
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# How to decompose a layered map?

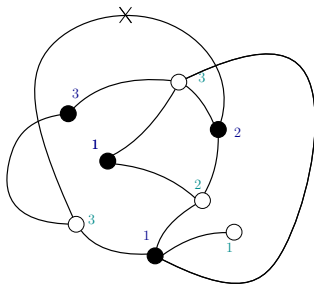


A non-orientable 3-layered map

Randomly (Dołęga–Féray–Śniady '14);

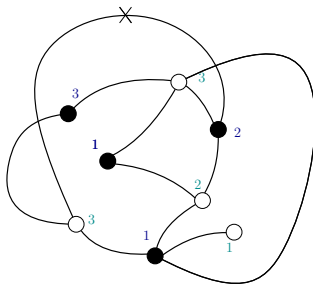
- works for  $k = 1$ .
- does not work for  $k \geq 3$ .

# Our decomposition algorithm



A non-orientable 3-layered map

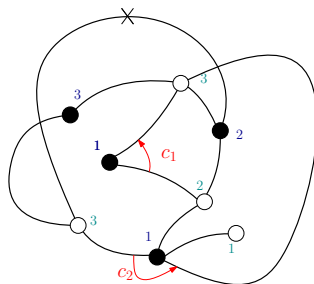
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A non-orientable 3-layered map

- We decompose the map in an increasing order of the layers.

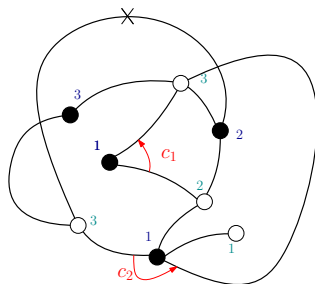
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Decomposition of a 3-layered map

- We decompose the map in an increasing order of the layers.
- We choose an order on the black vertices in layer 1, and a marked corner on each one of these vertices.

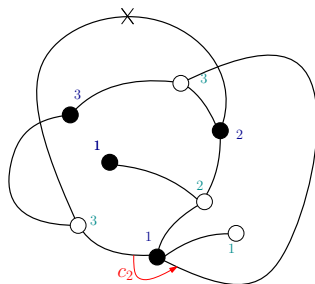
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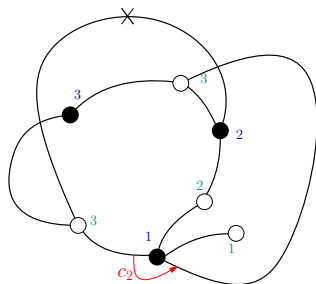
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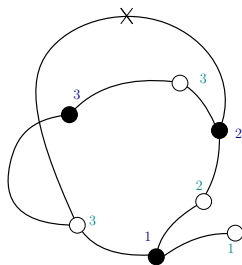
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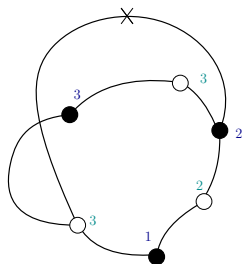


Decomposition of a 3-layered map

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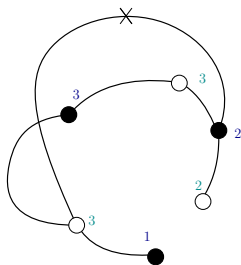
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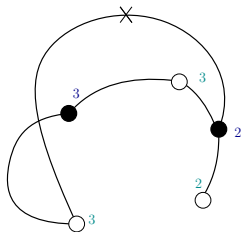
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# Shifted symmetric Jack polynomials

A function  $f(x_1, x_2, \dots)$  is  $\alpha$ -shifted symmetric if it is symmetric in the variables  $x_1, x_2 - 1/\alpha, x_3 - 2/\alpha, \dots$ .

$$f(\lambda) := f(\lambda_1, \dots, \lambda_{\ell(\lambda)}, 0, \dots).$$

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## Theorem (Knop–Sahi '96)

For any  $\mu$  there exists a unique  $\alpha$ -shifted symmetric function  $J_\mu^*$  such that

- $\deg(J_\mu^*) = |\mu|$ ,
- $J_\mu^*(\lambda) = 0$  if  $|\lambda| < |\mu|$ ,
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## Theorem (Lassalle '98)

For any  $\mu$  and  $\lambda$ , such that  $m := |\lambda| - |\mu| \geq 0$ . Then,

$$J_\mu^*(\lambda) = \left\langle J_\mu^{(\alpha)}, \frac{\partial^m}{\partial p_1^m} J_\lambda^{(\alpha)} \right\rangle.$$

# Characterization theorem

## Theorem (Féray '19)

*Fix a partition  $\mu$ . The Jack character  $\theta_\mu^{(\alpha)}$  is the unique  $\alpha$ -shifted symmetric function of degree  $|\mu|$  with top homogeneous part  $\alpha^{|\mu|-\ell(\mu)}/z_\mu \cdot p_\mu$ , such that  $\theta_\mu^{(\alpha)}(\lambda) = 0$  for any partition  $|\lambda| < |\mu|$ .*

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- We introduce the generating series of  $k$ -layered maps

$$F^{(k)}(t, \mathbf{p}, s_1, \dots, s_k) := \sum_{k\text{-layered maps } M} (-t)^{|\diamond(M)|} p_{\diamond(M)} \frac{b^{\vartheta(M)}}{2^{|\mathcal{V}_\bullet(M)|-cc(M)} \alpha^{cc(M)}} \prod_{1 \leq i \leq k} \frac{(-\alpha s_i)^{|\mathcal{V}_\circ^{(i)}(M)|}}{z_{\nu_\bullet^{(i)}(M)}}.$$



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We prove that this generating series satisfies the three conditions of the characterization theorem using differential operators.

# Idea of the proof

- The generating series of layered maps can be constructed using differential operators (Tutte decomposition):

$$F^{(k)}(t, \mathbf{p}, s_1, \dots, s_k) = \exp \left( \sum_{n \geq 1} \frac{(-t)^n}{n} B_n(\mathbf{p}, -\alpha s_1) \right) \cdot F^{(k-1)}(t, \mathbf{p}, s_2, \dots, s_k),$$

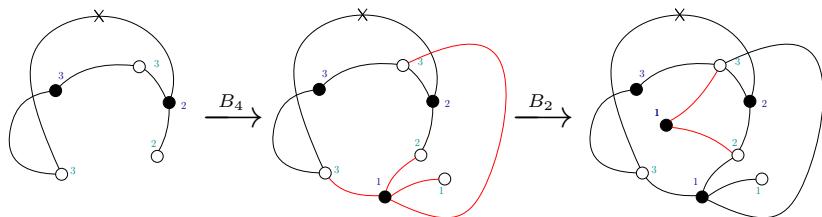
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In these operators  $B_n$ , we control:

- the face partition  $p_{\diamond(M)}$ ,
- the non-orientability weight  $b^{\vartheta(M)}$ ,
- the number of added white vertices.

# Differential operators and maps

Fix a bipartite map  $M$ .

$$\left( \sum_{i \geq 1} p_{i+1} \frac{i \partial}{\partial p_i} \right) p_{\diamond(M)} = \sum_e p_{\diamond(M \cup \{e\})}$$

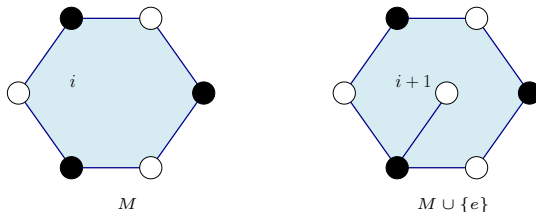
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# Differential operators and maps

$$\left( \sum_{i,j \geq 1} p_i p_j \frac{(i+j-1)\partial}{\partial p_{i+j-1}} + 2 \sum_{i,j \geq 1} p_{i+j+1} \frac{i\partial}{\partial p_i} \frac{j\partial}{\partial p_j} + \sum_{i \geq 1} p_{i+1} \frac{i\partial}{\partial p_i} \right) \cdot \frac{p_{\diamond(M)}}{\kappa(M)} \\ = \sum_e \frac{p_{\diamond(M \cup \{e\})}}{\kappa(M \cup \{e\})}$$

the sum is taken over all possible ways to add an edge  $e$  between two corners of  $M$ , and

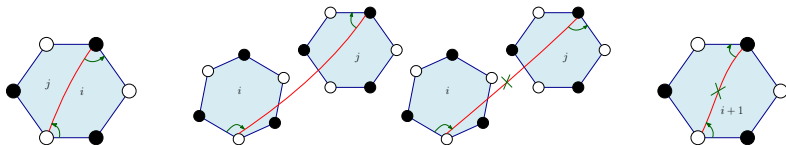
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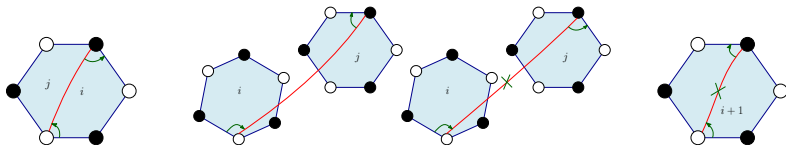


# Differential operators and maps

$$\left( \sum_{i,j \geq 1} p_i p_j \frac{(i+j-1)\partial}{\partial p_{i+j-1}} + (1+b) \sum_{i,j \geq 1} p_{i+j+1} \frac{i\partial}{\partial p_i} \frac{j\partial}{\partial p_j} + b \sum_{i \geq 1} p_{i+1} \frac{i\partial}{\partial p_i} \right) \cdot \frac{p_{\diamond(M)}}{\kappa(M)} \\ = \sum_e b^{\vartheta(M \cup \{e\}, e)} \frac{p_{\diamond(M \cup \{e\})}}{\kappa(M \cup \{e\})}$$

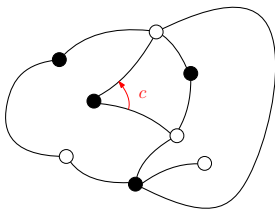
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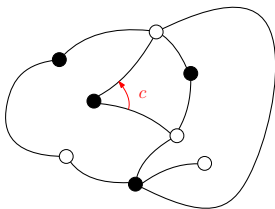
# Chapuy–Dołęga operators '22

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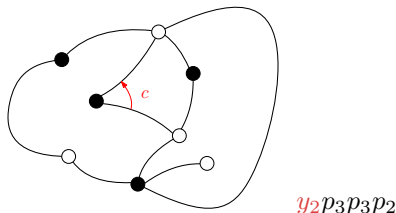
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We consider an additional alphabet  $Y := (y_0, y_1, y_2, \dots)$ . We associate to the root face a weight  $y_i$  and to other faces a weight  $p_i$ .

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# Chapuy–Dołęga operators

$$Y_+ = \sum_{i \geq 0} y_{i+1} \frac{\partial}{\partial y_i}, \quad \longrightarrow \text{adds a white leaf on the root corner.}$$

$$\Gamma_Y = \sum_{i,j \geq 1} y_i p_j \frac{\partial}{\partial y_{i+j-1}} + (1+b) \cdot \sum_{i,j \geq 1} y_{i+j} \frac{i \partial^2}{\partial p_i \partial y_{j-1}} + b \cdot \sum_{i \geq 1} y_{i+1} \frac{i \partial}{\partial y_i}$$

$\longrightarrow$  adds an edge between the root corner and a white corner of the map.

$$\Theta_Y := \sum_{i \geq 1} p_i \frac{\partial}{\partial y_i} \quad \longrightarrow \text{"forgets" the root.}$$

$$B_n(\mathbf{p}, u) := \Theta_Y (\Gamma_Y + u Y_+)^n \frac{y_0}{1+b} \quad \longrightarrow \text{adds a black vertex of degree } n.$$

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## Remark

The variables  $y_i$  are catalytic variables.

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$$\begin{aligned} F^{(k)}(t, \mathbf{p}, s_1, \dots, s_k) \\ = \exp \left( \sum_{n \geq 1} \frac{(-t)^n}{n} B_n(\mathbf{p}, -\alpha s_1) \right) \cdot F^{(k-1)}(t, \mathbf{p}, s_2, \dots, s_k), \end{aligned}$$

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**Recall:** we want to prove that  $F^{(k)}$  satisfies the properties of the characterization theorem:

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Fix a partition  $\mu$ . The Jack character  $\theta_\mu^{(\alpha)}$  is the unique  $\alpha$ -shifted symmetric function of degree  $|\mu|$  with top homogeneous part  $\alpha^{|\mu|-\ell(\mu)}/z_\mu \cdot p_\mu$ , such that  $\theta_\mu^{(\alpha)}(\lambda) = 0$  for any partition  $|\lambda| < |\mu|$ .



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We prove commutation relations

$$[B_n(\mathbf{p}, u), B_m(\mathbf{p}, u)] = 0, \text{ for } n, m \geq 1,$$

$$\left[ \sum_{n \geq 1} \frac{t^n}{n} B_n^>(\mathbf{p}, u), \sum_{n \geq 1} \frac{t^n}{n} B_n^>(\mathbf{p}, v) \right] = 0,$$

where

$$B_n^>(\mathbf{p}, u) := B_n(\mathbf{p}, u) - B_n(\mathbf{p}, 0).$$

# Application 1: Creation operators for Jack polynomials

## Theorem (BD–Dołęga '23+)

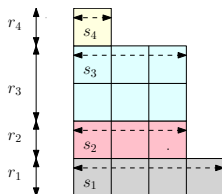
$$J_{(\lambda_1, \lambda_2, \dots, \lambda_\ell)}^{(\alpha)} = \mathcal{B}_{\lambda_1}^{(+)} \cdot \mathcal{B}_{\lambda_2}^{(+)} \cdots \mathcal{B}_{\lambda_\ell}^{(+)} \cdot 1,$$

where

$$\mathcal{B}_n^{(+)} := [t^n] \exp \left( \sum_{n \geq 1} \frac{(-t)^n}{n} B_n(\mathbf{p}, -\alpha n) \right)$$

# Application 2: Lassalle's conjecture 2008

## Stanley's coordinates of a Young diagram



The Young diagram of the partition  $[4, 3, 3, 1]$ , with  $\mathbf{s} = (4, 3, 3, 1)$  and  $\mathbf{r} = (1, 1, 2, 1)$  as Stanley coordinates.

## Theorem (Lassalle's conjecture on Jack characters)

*The normalized Jack character  $(-1)^{|\mu|} z_\mu \theta_\mu^{(\alpha)}$  is a polynomial in Stanley's coordinates  $r_1, r_2, \dots, -s_1, -s_2, \dots$ , and  $b$  with non-negative integer coefficients.*

### Proof:

- Non-negativity: the combinatorial interpretation in terms of maps.
- Integrality: (a different approach) operators from an integrable system of Nazarov–Sklyanin and combinatorics of Łukasiewicz paths.

# Towards a Macdonald generalization

## Theorem (BD–D’Adderio ’24+)

*There exists a family of creation operators  $(\Gamma_n^+)$ , such that for any partition  $\lambda$*

$$J_{\lambda}^{(q,t)} = \Gamma_{\lambda_1}^+ \Gamma_{\lambda_2}^+ \cdots \Gamma_{\lambda_{\ell(\lambda)}}^+ \cdot 1.$$

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We use these operators to introduce **Macdonald characters**;

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**Question:** Are these characters encoding a two parameter generalization of non-orientable maps?